

BELOZERSK, Russian Federation

WMO: 229390

Lat: 60.0203N

Lon: 37.7592E

Elev: 131

StdP: 99.77

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.6	(c) -25.0	(d) -31.4	(e) 0.2	(f) -28.5	(g) -27.7	(h) 0.3	(i) -24.9	(j) 10.4	(k) -4.7	(l) 9.7	(m) -4.9	(n) 1.5	(o) 170	(p) 0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.8	(c) 27.9	(d) 19.7	(e) 25.8	(f) 18.7	(g) 24.0	(h) 17.7	(i) 21.1	(j) 26.3	(k) 19.7	(l) 24.2	(m) 18.5	(n) 22.7	(o) 3.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.1	(b) 14.1	(c) 23.6	(d) 17.9	(e) 13.1	(f) 22.4	(g) 16.8	(h) 12.1	(i) 21.0	(j) 61.5	(k) 26.1	(l) 56.8	(m) 24.2	(n) 53.0	(o) 22.8	(p) 28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.6	7.5	-31.8	30.5	4.1	1.9	-34.7	31.8	-37.1	32.9	-39.5	34.0	-42.5	35.4
			-31.8	22.2	4.2	2.8	-34.8	24.2	-37.2	25.8	-39.6	27.4	-42.6	29.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-10.4	-8.7	-4.2	3.6	10.5	15.2	17.5	15.4	10.3	3.7	-2.6	-6.9		
	DBStd	11.00	7.25	7.60	5.24	4.56	5.18	4.07	3.61	3.53	3.76	3.87	5.65	7.12		
	HDD10.0	3006	633	523	442	201	59	4	1	2	41	198	377	525		
	HDD18.3	5421	891	757	700	443	248	112	59	104	241	454	627	783		
	CDD10.0	693	0	0	0	7	73	159	232	169	49	2	0	0		
	CDD18.3	67	0	0	0	0	4	17	33	13	1	0	0	0		
	CDH23.3	550	0	0	0	1	44	152	245	107	1	0	0	0		
	CDH26.7	122	0	0	0	0	7	29	59	27	0	0	0	0		
(14) Wind	WSAvg	3.4	3.5	3.8	3.9	3.5	3.4	3.1	2.6	2.9	3.1	3.5	3.7	3.9		
(15) Precipitation	PrecAvg	538	25	19	24	33	51	59	76	74	53	50	38	33		
	PrecMax	733	54	45	59	82	86	164	164	196	113	113	83	64		
	PrecMin	377	9	6	2	3	17	17	24	12	22	0	8	16		
	PrecStd	88	11	10	15	18	21	30	35	42	22	22	18	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	3.2	8.2	21.1	27.5	29.7	31.1	30.1	23.0	15.6	8.6	4.9		
		MCWB	1.1	1.7	3.6	12.8	16.6	21.0	21.9	21.8	16.8	12.1	7.7	4.1		
	2%	DB	1.2	1.9	5.3	16.8	24.0	27.1	28.1	25.8	20.1	12.1	6.3	3.0		
		MCWB	0.5	0.8	2.2	10.1	14.9	19.2	20.2	18.9	15.5	9.9	5.4	2.2		
	5%	DB	0.5	1.0	3.5	13.6	21.2	25.0	26.0	23.6	17.7	10.6	4.9	1.9		
		MCWB	0.0	0.1	1.1	7.9	13.7	18.0	19.8	18.0	14.3	9.0	4.4	1.2		
	10%	DB	-0.5	0.3	2.2	10.7	18.6	22.6	24.0	21.7	15.8	9.2	3.7	0.9		
		MCWB	-1.0	-0.4	0.5	6.1	12.5	16.9	18.3	16.9	13.1	7.9	3.2	0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	2.0	4.2	13.4	18.1	22.6	23.2	22.2	17.8	12.8	8.1	4.1		
		MCDB	1.7	2.9	6.9	19.9	25.2	27.8	27.9	27.6	21.7	14.8	8.4	4.8		
	2%	WB	0.7	0.9	2.7	10.6	16.1	20.3	21.3	20.0	16.1	10.7	5.6	2.4		
		MCDB	1.1	1.5	4.7	15.9	21.7	25.1	26.8	24.4	19.0	11.8	6.1	2.8		
	5%	WB	0.1	0.2	1.6	8.5	14.8	18.9	20.1	18.4	14.8	9.3	4.4	1.3		
		MCDB	0.5	0.8	3.0	12.3	19.2	23.3	24.8	22.6	17.2	10.4	4.9	1.7		
	10%	WB	-1.1	-0.4	0.8	6.5	13.4	17.5	19.0	17.3	13.4	7.9	3.3	0.4		
		MCDB	-0.5	0.3	2.1	10.2	17.5	21.6	23.1	21.1	15.5	9.1	3.7	0.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.0	7.6	8.7	9.5	9.0	8.8	8.5	6.8	4.7	3.8	4.6		
		MCDBR	3.7	4.4	7.9	13.2	13.6	12.2	11.7	11.8	9.9	6.4	4.2	3.3		
	5% WB	MCWBR	3.8	4.1	5.5	7.8	6.2	5.9	5.6	5.7	5.6	4.8	3.9	3.2		
		MCWBR	3.7	4.3	5.9	12.0	10.8	10.9	10.6	10.6	8.6	5.8	4.1	3.3		
(40) Clear-Sky Solar Irradiance	taub		0.262	0.306	0.328	0.370	0.358	0.356	0.389	0.382	0.346	0.327	0.286	0.240		
		taud	2.054	2.069	2.126	2.248	2.430	2.460	2.382	2.383	2.464	2.390	2.155	2.011		
	Ebn at Noon		487	663	786	825	870	876	832	808	781	665	457	357		
		Edh at Noon	46	81	105	113	102	100	106	99	77	60	42	31		
(44) All-Sky Solar Radiation	RadAvg		0.32	0.98	2.40	3.83	5.07	5.35	5.19	3.98	2.31	0.97	0.30	0.17		
	RadStd		0.04	0.22	0.33	0.46	0.33	0.32	0.64	0.27	0.23	0.16	0.05	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.69	N/A	N/A	N/A	N/A	N/A	-200	-244	N/A	N/A

Nomenclature: See separate page